Metric Mathematical Derivations for EquiMed-DSS

Formulae Matched to the EquiMed-DSS Library Implementation

John Weirstrass Muteba Mwamba
School of Health Policy and Management, York University, Toronto, Canada
Developer and Modeler of EquiMed-DSS
jwmm@yorku.ca

EquiMed-DSS version 1.9.5

Purpose and Verification Scope

This technical document derives the mathematical definitions used by the EquiMed-DSS library, version 1.9.5. The formulae below are aligned with the local implementation in the package source code, especially the modules `domain1`, `domain2`, `domain3`, `domain4`, `domain5`, `geographic`, `appendix`, and `statistics`. Where an earlier derivation document used a more general or theoretical expression, the implementation-specific expression is reported here as the authoritative formula.

The library documentation lists 37 named metrics. This document gives 39 derivations by adding two implemented statistical estimands used with the metric suite: hierarchical variance partitioning and mediation/proportion mediated. These two estimands are implemented in `equimed_dss.statistics` and are included because they are used to interpret fairness and governance results in the EquiMed-DSS framework.

Each metric is numbered in the overview table and in its subsection title. Some implemented classes return companion quantities, for example HER with Bias-Gini or CPS with CFU. These companion quantities use the parent metric number with a letter suffix. Every displayed equation is written in a numbered LaTeX `equation` or `align` environment so it can be referenced directly after compilation.

Notation

Let $i = 1, \dots, n$ index observations, $g \in \mathcal{G}$ index demographic or intersectional groups, $s \in \mathcal{S}$ index healthcare-system strata, and $r \in \mathcal{R}$ index geographic regions. Let $\hat{Y}_i$ denote a model prediction, Y_i the corresponding reference outcome, C_i a confidence score, and $Z_i = \mathbb{I}(\hat{Y}_i = Y_i)$ a correctness indicator. For text outputs, let R_i denote a response, $D(R_i)$ the set of diagnoses mentioned in the response, and $D^\star$ a reference differential diagnosis set. Unless otherwise stated, all means are empirical means over the supplied input arrays.

No.	Metric	Abbreviation	Canonical implementation
1	Inter-rater reliability	ICC(2,1)	`domain1.icc`
2	Embedding consistency score	ECS	`domain1.ecs`
3	Decision flip rate	DFR	`domain1.dfr`

No.	Metric	Abbreviation	Canonical implementation
4	Hierarchical equity ratio and Bias-Gini	HER	`domain2.her`
5	Harm-adjusted fairness gap	HAFG	`domain2.hafg`
6	Ethical risk index and safety violation rate	ERI	`domain2.eri`
7	Intersectional bias score	IBS	`domain2.ibs`
8	Temporal fairness drift	TFD	`domain3.tfd`
9	Audit traceability score	ATS	`domain3.ats`
10	Governance compliance index	GCI	`domain3.gci`
11	Semantic parity gap	SPG	`domain4.spg`
12	Clinical hallucination rate	CHR	`domain4.chr`
13	Instructional vulnerability index	IVI	`domain4.ivi`
14	Geographic representation index and geographic bias	GRI	`domain4.gri`
15	Intersectional calibration error	ICE	`domain5.calibration`
16	Weighted clinical harm-adjusted fairness gap	wHAFG	`domain5.harm`
17	Counterfactual parity score and counterfactual unfairness	CPS	`domain5.counterfactual`
18	Semantic robustness parity index	SRPI	`domain5.counterfactual`
19	Lexical diversity disparity index	LDDI	`domain5.text`
20	Recommendation entropy gap	REG	`domain5.text`
21	Clinical information density ratio	CIDR	`domain5.text`
22	Diagnostic completeness index	DCI	`domain5.text`
23	Uncertainty quantification gap	UQG	`domain5.text`
24	Geographic representation bias index	GRBI	`domain5.geographic_bias`
25	Healthcare system stratified fairness	HSSF	`domain5.system`
26	Intersectional Shapley fairness value	ISFV	`domain5.shapley`
27	Burden-evidence mismatch index	BEMI	`geographic.burden_evidence`
28	Geographic concentration of coverage	GCC	`geographic.concentration`
29	Bootstrap confidence interval	BCI	`appendix.advanced_metrics`
30	Statistical power analysis	SPA	`appendix.advanced_metrics`
31	Bias concentration index	BCI-bias	`appendix.advanced_metrics`
32	Mutual information content	MIC	`appendix.advanced_metrics`
33	Jensen-Shannon divergence	JSD	`appendix.advanced_metrics`
34	Wasserstein distance	WD	`appendix.advanced_metrics`
35	Network modularity	NM	`appendix.advanced_metrics`
36	Transparency score	TS	`appendix.advanced_metrics`

No.	Metric	Abbreviation	Canonical implementation
37	Robustness certification score	RCS	`appendix.advanced_metrics`
38	Hierarchical variance partitioning	HLM/VPC	`statistics.hierarchical`
39	Causal mediation and proportion mediated	PM	`statistics.mediation`

1 Domain 1: Reliability and Robustness

1.1 Metric 1: Inter-rater Reliability, ICC(2,1)

Let X_{ij} be the score assigned to item i by judge j, with n items and k judges. Define the grand mean $\bar{X}_{..}$, item means $\bar{X}_{i.}$, and judge means $\bar{X}_{.j}$. This follows the Shrout-Fleiss ICC family and the Bland-Altman agreement convention [1, 2]. The implementation computes

$$SS_{\text{items}} = k\sum_{i=1}^{n}(\bar{X}_{i.} - \bar{X}_{..})^2, \tag{1}$$

$$SS_{\text{judges}} = n\sum_{j=1}^{k}(\bar{X}_{.j} - \bar{X}_{..})^2, \tag{2}$$

$$SS_{\text{error}} = \sum_{i=1}^{n}\sum_{j=1}^{k}(X_{ij} - \bar{X}_{..})^2 - SS_{\text{items}} - SS_{\text{judges}}. \tag{3}$$

The mean squares are

$$MS_R = \frac{SS_{\text{items}}}{n-1}, \qquad MS_C = \frac{SS_{\text{judges}}}{k-1}, \qquad MS_E = \frac{SS_{\text{error}}}{(n-1)(k-1)}. \tag{4}$$

EquiMed-DSS implements the two-way random-effects, single-measure, absolute agreement intraclass correlation coefficient as

$$ICC(2,1) = \frac{MS_R - MS_E}{MS_R + (k-1)MS_E + \frac{k}{n}(MS_C - MS_E)}. \tag{5}$$

The same class also computes Bland-Altman pairwise agreement. For two judges a and b,

$$d_i = X_{ia} - X_{ib}, \tag{6}$$

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \tag{7}$$

$$s_d = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2}, \tag{8}$$

$$LOA_{\text{lower}}, LOA_{\text{upper}} = \bar{d} \pm 1.96 s_d. \tag{9}$$

95% CI (this section). $ICC(2,1)$ is reported with a percentile bootstrap over targets $\times$ judges (items): resample with replacement $B = 1000$ times, recompute $ICC(2,1)$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{10}$$

1.2 Metric 2: Embedding Consistency Score

For paired original and perturbed embedding vectors e_i and $\tilde{e}_i$, the library calculates cosine distance, not cosine similarity:

$$ECS_i = 1 - \frac{e_i^\top \tilde{e}_i}{\|e_i\|_2 \|\tilde{e}_i\|_2}. \tag{11}$$

If either vector has zero norm, the implementation sets the cosine similarity to zero, so $ECS_i = 1$. The reported summary statistics are

$$\overline{ECS} = \frac{1}{n}\sum_{i=1}^{n} ECS_i, \tag{12}$$

$$SD(ECS) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(ECS_i - \overline{ECS})^2}, \tag{13}$$

$$\widetilde{ECS} = \text{median}(ECS_1, \dots, ECS_n). \tag{14}$$

Lower values indicate greater embedding stability.

95% CI (this section). ECS is reported with a percentile bootstrap over embedding pairs: resample with replacement $B = 1000$ times, recompute ECS, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{15}$$

1.3 Metric 3: Decision Flip Rate

Let A_i be the model decision on the original input and B_i the decision on the paired counterfactual or perturbed input. EquiMed-DSS defines

$$DFR = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(A_i \neq B_i). \tag{16}$$

If $x = \sum_i \mathbb{I}(A_i \neq B_i)$ and $\hat{p} = x/n$, the Wilson interval returned by the library is

$$d = 1 + \frac{z^2}{n}, \tag{17}$$

$$c = \frac{\hat{p} + \frac{z^2}{2n}}{d}, \tag{18}$$

$$h = \frac{z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{d}, \tag{19}$$

$$CI_{95} = [\max(0, c-h), \min(1, c+h)], \tag{20}$$

with $z = 1.96$, following Wilson's score interval for binomial proportions [3].

2 Domain 2: Fairness, Equity, and Ethics

95% CI (this section). As a binomial proportion, DFR is reported with a Wilson 95% score interval over decisions (Metric 9 form), with a one-sided score test against a tolerated flip rate.

2.1 Metric 4: Hierarchical Equity Ratio

Let q_g be a group-specific performance score and q_0 the reference-group score. The implementation calculates

$$HER_g = \frac{q_g}{q_0}, \tag{21}$$

with $HER_g = 0$ if $q_0 = 0$. Values in $[0.8, 1.25]$ are labelled equitable by the implemented four-fifths-rule convention [4].

95% CI (this section). HER is reported with a percentile bootstrap of the gap over per-group observations: resample with replacement $B = 1000$ times, recompute HER, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{22}$$

(needs observation-level group scores; otherwise reported unavailable).

2.2 Metric 4a: Bias-Gini Dispersion

For group scores $q_1, \ldots, q_K$ with mean $\bar{q}$, the implemented dispersion index is the standard Gini coefficient [5]:

$$G_{\text{bias}} = \frac{\sum_{i=1}^{K} \sum_{j=1}^{K} |q_i - q_j|}{2K^2\bar{q}}. \tag{23}$$

If the score list is empty or $\bar{q} = 0$, the function returns zero.

95% CI (this section). Bias-Gini is reported with a percentile bootstrap over group scores: resample with replacement $B = 1000$ times, recompute Bias-Gini, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{24}$$

2.3 Metric 5: Harm-adjusted Fairness Gap

For two groups, let FN_g and FP_g be false-negative and false-positive counts. With default costs $c_{FN} = 10$ and $c_{FP} = 3$, group harm is

$$H_g = c_{FN} FN_g + c_{FP} FP_g. \tag{25}$$

EquiMed-DSS reports the absolute gap

$$\Delta_H = |H_1 - H_2| \tag{26}$$

and the normalized harm-adjusted fairness gap

$$HAFG = \frac{|H_1 - H_2|}{\max(H_1, H_2)}. \tag{27}$$

If both group harms are zero, the denominator is zero and the implemented value is 0.

95% CI (this section). $HAFG$ is reported with a percentile bootstrap of the gap over per-case error labels: resample with replacement $B = 1000$ times, recompute $HAFG$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{28}$$

(needs observation-level cases; otherwise reported unavailable).

2.4 Metric 6: Ethical Risk Index

Let $v = 1, \ldots, V$ index detected ethical or safety violations and let s_v be their severity scores. For N total model outputs,

$$ERI = \frac{\sum_{v=1}^{V} s_v}{N}. \tag{29}$$

If $N = 0$, the function returns zero.

95% CI (this section). ERI is reported with a percentile bootstrap over the per-output severity vector (severity for violations, 0 otherwise): resample with replacement $B = 1000$ times, recompute ERI, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{30}$$

2.5 Metric 6a: Safety Violation Rate

In the same implementation as ERI, the safety violation rate is

$$SVR = 1000\frac{V}{N}, \tag{31}$$

that is, violations per 1000 model outputs.

95% CI (this section). As a binomial proportion, SVR is reported with a Wilson 95% score interval over outputs (Metric 9 form).

2.6 Metric 7: Intersectional Bias Score

Let $u_g \in \mathbb{R}^d$ be a vector of metrics for subgroup g. The implementation calculates Euclidean distances

$$d(g, h) = \|u_g - u_h\|_2 \tag{32}$$

and converts them to similarities by inverse distance:

$$S(g, h) = \frac{1}{1 + d(g, h)}. \tag{33}$$

The outlier subgroup is the subgroup with the largest mean distance over the full distance row, including the zero self-distance used by the implementation:

$$g^\star = \arg\max_g \frac{1}{K} \sum_{h \in \mathcal{G}} d(g, h). \tag{34}$$

The same class also computes simplified eta-squared-style interaction effects. For a categorical attribute A, with grand mean $\bar{Y}$ and group means $\bar{Y}_a$, the main-effect proxy is

$$\eta_A^2 = \frac{\sum_a n_a (\bar{Y}_a - \bar{Y})^2}{\sum_i (Y_i - \bar{Y})^2}. \tag{35}$$

For the race-by-gender interaction, the implementation subtracts the race and gender main-effect proxies from the combined race-gender proxy.

3 Domain 3: Governance and Transparency

95% CI (this section). IBS is reported with a percentile bootstrap over metric dimensions of the subgroup vectors: resample with replacement $B = 1000$ times, recompute IBS, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{36}$$

3.1 Metric 8: Temporal Fairness Drift

For a time series of fairness metric values $m_1, \ldots, m_T$, EquiMed-DSS computes

$$\bar{m} = \frac{1}{T}\sum_{t=1}^{T} m_t, \tag{37}$$

$$s_m = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(m_t - \bar{m})^2}. \tag{38}$$

The three-sigma control limits are

$$UCL = \bar{m} + 3s_m, \tag{39}$$

$$LCL = \bar{m} - 3s_m. \tag{40}$$

A drift point is flagged when $m_t > UCL$ or $m_t < LCL$, following the Shewhart-style control-chart logic used in statistical process control [6].

95% CI (this section). TFD is reported with a percentile bootstrap over the observed time series: resample with replacement $B = 1000$ times, recompute TFD, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{41}$$

3.2 Metric 9: Audit Traceability Score

Let x be the number of traceable decisions among n audited decisions. The score is

$$ATS = \frac{x}{n}. \tag{42}$$

The implementation returns a Wilson-style shrinkage interval using

$$z = 1.96, \tag{43}$$

$$\tilde{p} = \frac{x + z^2/2}{n + z^2}, \tag{44}$$

$$SE_{\tilde{p}} = \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n + z^2}}, \tag{45}$$

$$CI_{95} = [\max(0, \tilde{p} - zSE_{\tilde{p}}), \min(1, \tilde{p} + zSE_{\tilde{p}})]. \tag{46}$$

The interval uses Wilson score shrinkage for a binomial proportion [3]. The score is labelled compliant at $ATS \geq 0.95$.

3.3 Metric 10: Governance Compliance Index

Let M be the number of mandated policies and E the number evaluated as enforced. The implementation defines

$$GCI = \frac{E}{M}. \tag{47}$$

When no policies are provided, the returned value is zero.

4 Domain 4: Representation and Robustness

95% CI (this section). As a binomial proportion, GCI is reported with a Wilson 95% score interval over checklist items (Metric 9 form).

4.1 Metric 11: Semantic Parity Gap

Let $P = \{p_i\}_{i=1}^{n_p}$ be embeddings for privileged-group prompts and $M = \{m_j\}_{j=1}^{n_m}$ embeddings for matched marginalized-group prompts. Their centroids are

$$\bar{p} = \frac{1}{n_p}\sum_{i=1}^{n_p} p_i, \qquad \bar{m} = \frac{1}{n_m}\sum_{j=1}^{n_m} m_j. \tag{48}$$

The Euclidean SPG reported by the implementation is

$$SPG_{\text{Euc}} = \left\|\bar{p} - \bar{m}\right\|_2. \tag{49}$$

The cosine variant is

$$SPG_{\cos} = 1 - \frac{\bar{p}^\top \bar{m}}{\left\|\bar{p}\right\|_2 \left\|\bar{m}\right\|_2}, \tag{50}$$

with value zero if the denominator is zero.

95% CI (this section). SPG is reported with a percentile bootstrap over embedding rows within each group (two-sample bootstrap): resample with replacement $B = 1000$ times, recompute SPG, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{51}$$

4.2 Metric 12: Clinical Hallucination Rate

Let $c = 1, \ldots, C$ index extracted clinical claims and $S(c, K) \in [0, 1]$ be a precomputed support score against retrieved context K. With entailment threshold τ, the implemented hallucination rate is

$$CHR = \frac{1}{C}\sum_{c=1}^{C} \mathbb{I}\{S(c, K) < \tau\}. \tag{52}$$

With optional severity weights w_c,

$$CHR_w = \frac{\sum_{c=1}^{C} w_c \mathbb{I}\{S(c, K) < \tau\}}{\sum_{c=1}^{C} w_c}. \tag{53}$$

If no weights are supplied, the implementation sets $CHR_w = CHR$.

95% CI (this section). As a binomial proportion, CHR is reported with a Wilson 95% score interval over claims (Metric 9 form), with a one-sided score test against a tolerated rate.

4.3 Metric 13: Instructional Vulnerability Index

Let A_i be the neutral-output decision and B_i the paired biased-instruction decision for the same case. The flip component is

$$IVI = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(A_i \neq B_i). \tag{54}$$

If the outputs can be coerced to numeric values, the implementation also returns the directional effect

$$IVI_{\text{effect}} = \frac{1}{n}\sum_{i=1}^{n} B_i - \frac{1}{n}\sum_{i=1}^{n} A_i. \tag{55}$$

The paired-counterfactual framing is conceptually related to counterfactual fairness, although IVI targets prompt framing rather than protected-attribute interventions [8].

95% CI (this section). As a binomial proportion, IVI is reported with a Wilson 95% score interval over case pairs (Metric 9 form), with a one-sided score test against a tolerated rate.

4.4 Metric 14: Geographic Representation Index

Let L be the set of unique locations represented in a corpus and let $W \subseteq L$ be those counted as Western or high-income. The implemented index is set-based:

$$GRI = \frac{|L| - |W|}{|L|}. \tag{56}$$

Duplicates do not change the score.

95% CI (this section). GRI is reported with a percentile bootstrap over location mentions: resample with replacement $B = 1000$ times, recompute GRI, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^{*}_{(0.025)},\ \hat{\theta}^{*}_{(0.975)}\right]. \tag{57}$$

4.5 Metric 14a: Geographic Bias Correlation

For paired values (x_i, y_i), where x_i is a per-query GRI value and y_i is the corresponding non-Western error rate, the library returns either Pearson or Spearman correlation. For Pearson,

$$GB = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}. \tag{58}$$

5 Domain 5: Technical-supplement Fairness

95% CI (this section). GB is reported with a percentile bootstrap over paired (GRI, error) points; Fisher-z or bootstrap: resample with replacement $B = 1000$ times, recompute GB, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^{*}_{(0.025)},\ \hat{\theta}^{*}_{(0.975)}\right]. \tag{59}$$

5.1 Metric 15: Intersectional Calibration Error

For group g and bin b, let S_{gb} be samples in intersectional group g whose confidence falls in bin b. The implementation uses equal-width bins on $[0, 1]$. This extends expected calibration error as used in neural-network calibration studies [7]. Let

$$acc(S_{gb}) = \frac{1}{|S_{gb}|} \sum_{i \in S_{gb}} Z_i, \tag{60}$$

$$conf(S_{gb}) = \frac{1}{|S_{gb}|} \sum_{i \in S_{gb}} C_i. \tag{61}$$

The group-specific calibration error is

$$ECE_g = \sum_b \frac{|S_{gb}|}{|S_g|} \left| acc(S_{gb}) - conf(S_{gb}) \right|. \tag{62}$$

The intersectional calibration error is the population-weighted average

$$ICE = \sum_g \frac{|S_g|}{\sum_h |S_h|} ECE_g. \tag{63}$$

The maximum calibration gap returned as `delta_ice` is

$$\Delta ICE = \max_g ECE_g - \min_g ECE_g. \tag{64}$$

95% CI (this section). ICE is reported with a percentile bootstrap over samples (group, confidence, correctness): resample with replacement $B = 1000$ times, recompute ICE, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)} \right]. \tag{65}$$

5.2 Metric 16: Weighted Clinical Harm-adjusted Fairness Gap

Let w_i be a clinical severity weight and $\ell_i = L(\hat{Y}_i, Y_i)$ be a per-sample loss. For group g,

$$H(g) = \frac{1}{n_g} \sum_{i: G_i = g} w_i \ell_i. \tag{66}$$

The implemented maximum gap is

$$wHAFG_{\max} = \max_g H(g) - \min_g H(g). \tag{67}$$

95% CI (this section). $wHAFG$ is reported with a percentile bootstrap of the gap over samples: resample with replacement $B = 1000$ times, recompute $wHAFG$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)} \right]. \tag{68}$$

5.3 Metric 17: Counterfactual Parity Score

Let $s_i \in [0, 1]$ be a precomputed semantic similarity between the original response and the response under a demographic swap. For a single pair type,

$$CPS = \frac{1}{n} \sum_{i=1}^{n} s_i. \tag{69}$$

The demographic-swap construction is grounded in the counterfactual fairness intuition that decisions should be stable under protected-attribute changes when clinically relevant facts are unchanged [8]. For multiple swap-pair labels p, the library calculates

$$CPS_p = \frac{1}{n_p} \sum_{i \in p} s_i, \qquad CPS = \frac{1}{\sum_p n_p} \sum_p \sum_{i \in p} s_i. \tag{70}$$

95% CI (this section). CPS is reported with a percentile bootstrap over counterfactual pairs: resample with replacement $B = 1000$ times, recompute CPS, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)} \right]. \tag{71}$$

5.4 Metric 17a: Counterfactual Unfairness

The implementation returns a complement to CPS. For a single pair,

$$CFU = 1 - CPS. \tag{72}$$

For multiple pairs,

$$CFU = 1 - \min_p CPS_p. \tag{73}$$

95% CI (this section). As a binomial proportion, CFU is reported with a Wilson 95% score interval over pairs / 1 − CPS (Metric 9 form).

5.5 Metric 18: Semantic Robustness Parity Index

Let R_i be a per-query robustness score, usually the mean similarity among responses to semantically equivalent paraphrases. For group g,

$$R(g) = \frac{1}{n_g} \sum_{i: G_i = g} R_i. \tag{74}$$

The implemented parity ratio is

$$SRPI = \frac{\min_g R(g)}{\max_g R(g)}. \tag{75}$$

If the maximum group robustness is zero, the function returns zero.

95% CI (this section). $SRPI$ is reported with a percentile bootstrap over pooled per-query robustness scores (tagged by group): resample with replacement $B = 1000$ times, recompute $SRPI$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)} \right]. \tag{76}$$

5.6 Metric 19: Lexical Diversity Disparity Index

Let T_g be the multiset of tokens pooled across all responses in group g and V_g its vocabulary. The implemented root type-token ratio is

$$RTTR(g) = \frac{|V_g|}{\sqrt{|T_g|}}. \tag{77}$$

The disparity index is

$$LDDI = \max_g RTTR(g) - \min_g RTTR(g). \tag{78}$$

With $RTTR_{\text{all}}$ computed after pooling all groups,

$$LDDI_{\text{norm}} = \frac{LDDI}{RTTR_{\text{all}}}. \tag{79}$$

95% CI (this section). $LDDI$ is reported with a percentile bootstrap over pooled group responses: resample with replacement $B = 1000$ times, recompute $LDDI$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{80}$$

5.7 Metric 20: Recommendation Entropy Gap

Let $P_g(t)$ be the empirical distribution of recommendation labels t in group g. The group recommendation entropy is based on Shannon entropy [9]:

$$H(T \mid g) = -\sum_t P_g(t) \log_2 P_g(t). \tag{81}$$

The implemented gap is

$$REG = \max_g H(T \mid g) - \min_g H(T \mid g). \tag{82}$$

The implementation also returns

$$REG_{KL} = \max_g \sum_t P_g(t) \log_2 \frac{P_g(t)}{P(t)}, \tag{83}$$

where $P(t)$ is the marginal recommendation distribution.

95% CI (this section). REG is reported with a percentile bootstrap over pooled group recommendations: resample with replacement $B = 1000$ times, recompute REG, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{84}$$

5.8 Metric 21: Clinical Information Density Ratio

For response i, let a_i be the number of extracted clinical concepts and b_i the number of tokens. The response-level clinical information density is

$$CID_i = 100\frac{a_i}{b_i}. \tag{85}$$

For group g,

$$CID(g) = \frac{1}{n_g} \sum_{i:G_i=g} CID_i. \tag{86}$$

The group ratio is

$$CIDR(g) = \frac{CID(g)}{\max_h CID(h)}, \tag{87}$$

and the reported parity summary is

$$CIDR_{\min} = \min_g CIDR(g). \tag{88}$$

If the maximum group density is zero, all implemented ratios are set to zero.

95% CI (this section). $CIDR$ is reported with a percentile bootstrap over pooled (concepts, tokens) pairs (tagged by group): resample with replacement $B = 1000$ times, recompute $CIDR$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{89}$$

5.9 Metric 22: Diagnostic Completeness Index

Let $D^\star$ be the reference differential diagnosis set. For response i,

$$DCI_i = \frac{|D(R_i) \cap D^\star|}{|D^\star|}. \tag{90}$$

The group mean is

$$DCI(g) = \frac{1}{n_g} \sum_{i:G_i=g} DCI_i, \tag{91}$$

and the implemented disparity is

$$\Delta DCI = \max_g DCI(g) - \min_g DCI(g). \tag{92}$$

When diagnosis weights w_d are supplied, the weighted response score is

$$wDCI_i = \frac{\sum_{d \in D(R_i) \cap D^\star} w_d}{\sum_{d \in D^\star} w_d}. \tag{93}$$

95% CI (this section). DCI is reported with a percentile bootstrap over pooled group responses: resample with replacement $B = 1000$ times, recompute DCI, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{94}$$

5.10 Metric 23: Uncertainty Quantification Gap

Let h_i be the number of hedging terms in response R_i and q_i the number of sentences. The uncertainty density is

$$UD_i = \frac{h_i}{q_i}. \tag{95}$$

If no sentence is detected, $UD_i = 0$. For group g,

$$UD(g) = \frac{1}{n_g} \sum_{i:G_i=g} UD_i. \tag{96}$$

The implemented uncertainty quantification gap is

$$UQG = \max_g UD(g) - \min_g UD(g). \tag{97}$$

95% CI (this section). UQG is reported with a percentile bootstrap over pooled group responses: resample with replacement $B = 1000$ times, recompute UQG, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{98}$$

5.11 Metric 24: Geographic Representation Bias Index

Let $p_c(r)$ be the normalized corpus evidence share in region r and $p_b(r)$ the normalized burden share. EquiMed-DSS implements directed KL divergence in nats [10]:

$$GRBI = D_{\mathrm{KL}}(P_c \| P_b) = \sum_{r:p_c(r)>0} p_c(r) \log \frac{p_c(r)}{p_b(r)}. \tag{99}$$

If $p_c(r) > 0$ and $p_b(r) = 0$, the implementation raises an error because KL is undefined. With optional high-income regions $\mathcal{H}$, the high-income overrepresentation ratio is

$$HIC_{\mathrm{ratio}} = \frac{\sum_{r\in\mathcal{H}} p_c(r)}{\sum_{r\in\mathcal{H}} p_b(r)}. \tag{100}$$

95% CI (this section). $GRBI$ is reported with a percentile bootstrap over evidence records: resample with replacement $B = 1000$ times, recompute $GRBI$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{101}$$

(needs record-level input; otherwise reported unavailable).

5.12 Metric 25: Healthcare System Stratified Fairness

For system stratum s, let

$$\Delta_s = \max_g \mathbb{E}[Y \mid G = g, S = s] - \min_g \mathbb{E}[Y \mid G = g, S = s]. \tag{102}$$

The implemented within-system fairness gap is

$$HSSF = \sum_s P(S = s)\Delta_s. \tag{103}$$

The implementation returns $\Delta_{\text{within}} = HSSF$. It also returns the population-weighted between-system variance

$$\Delta_{\text{between}} = \sum_s P(S = s) \left(\mathbb{E}[Y \mid S = s] - \sum_u P(S = u)\mathbb{E}[Y \mid S = u] \right)^2 . \tag{104}$$

95% CI (this section). $HSSF$ is reported with a percentile bootstrap of the gap over samples: resample with replacement $B = 1000$ times, recompute $HSSF$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)} \right] . \tag{105}$$

5.13 Metric 26: Intersectional Shapley Fairness Value

Let $\mathcal{A} = \{A_1, \dots, A_m\}$ be protected attributes. The attribution formula follows Shapley's cooperative-game value [19]. For a subset $S \subseteq \mathcal{A}$, the implemented characteristic function is

$$v(S) = \max_{a \in \text{dom}(S)} \mathbb{E}[Y \mid A_S = a] - \min_{a \in \text{dom}(S)} \mathbb{E}[Y \mid A_S = a], \tag{106}$$

with $v(\varnothing) = 0$. Cells below the configured minimum cell size are ignored. The Shapley attribution for attribute A_j is

$$\phi_j = \sum_{S \subseteq \mathcal{A} \setminus \{A_j\}} \frac{|S|!(m - |S| - 1)!}{m!} \left[v(S \cup \{A_j\}) - v(S) \right] . \tag{107}$$

The total disparity is $v(\mathcal{A})$. Pairwise interactions are

$$I(A_j, A_k) = v(\{A_j, A_k\}) - v(\{A_j\}) - v(\{A_k\}). \tag{108}$$

6 Geographic Module

95% CI (this section). $ISFV$ is reported with a percentile bootstrap over samples (CI on the total disparity $v(A)$): resample with replacement $B = 1000$ times, recompute $ISFV$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)} \right] . \tag{109}$$

6.1 Metric 27: Burden-evidence Mismatch Index

Let $e(r)$ be the normalized evidence share and $b(r)$ the normalized burden share over the union of supplied regions. EquiMed-DSS defines BEMI as total variation distance [12]:

$$BEMI = \frac{1}{2} \sum_{r \in \mathcal{R}} |e(r) - b(r)|. \tag{110}$$

The per-region mismatch and ratio returned by the implementation are

$$M(r) = e(r) - b(r), \tag{111}$$

$$\rho(r) = \frac{e(r)}{b(r)}. \tag{112}$$

The ratio $\rho(r)$ is finite only when $b(r) > 0$; the implementation records a missing value when the burden share is zero. The most underserved region is the region with the minimum $M(r)$.

95% CI (this section). $BEMI$ is reported with a percentile bootstrap over geolocated evidence records: resample with replacement $B = 1000$ times, recompute $BEMI$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{113}$$

(needs record-level input; otherwise reported unavailable).

6.2 Metric 28: Geographic Concentration of Coverage

Let $x_r \geq 0$ be regional evidence counts for R regions and $p_r = x_r / \sum_u x_u$. The raw categorical Gini is

$$G_{\text{raw}} = \frac{\sum_{r=1}^{R} \sum_{u=1}^{R} |x_r - x_u|}{2R \sum_{r=1}^{R} x_r}. \tag{114}$$

Because the maximum raw Gini for R categories is $(R-1)/R$, the implementation uses the corrected value

$$G^\star = \frac{R}{R-1} G_{\text{raw}}. \tag{115}$$

The normalized Shannon entropy follows Shannon's entropy definition [9]:

$$H_{\text{norm}} = -\frac{\sum_{r:p_r>0} p_r \log p_r}{\log R}, \tag{116}$$

and the concentration score is

$$C_{\text{geo}} = 1 - H_{\text{norm}}. \tag{117}$$

7 Appendix Metrics

95% CI (this section). GCC is reported with a percentile bootstrap over geolocated evidence records: resample with replacement $B = 1000$ times, recompute GCC, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{118}$$

(needs record-level input; otherwise reported unavailable).

7.1 Metric 29: Bootstrap Confidence Interval

Given data $x_1, \ldots, x_n$ and a statistic $T(\cdot)$, EquiMed-DSS draws B bootstrap resamples x^{*b} of size n with replacement and computes

$$\theta^*_b = T(x^{*b}), \qquad b = 1, \ldots, B. \tag{119}$$

For significance level α, the implemented percentile interval is

$$CI_{1-\alpha} = \left[Q_{\alpha/2}(\theta^*_1, \ldots, \theta^*_B), Q_{1-\alpha/2}(\theta^*_1, \ldots, \theta^*_B)\right]. \tag{120}$$

The observed statistic is $T(x_1, \ldots, x_n)$. The percentile construction follows the nonparametric bootstrap framework of Efron and Tibshirani [13, 14].

95% CI (this section). this metric IS the percentile bootstrap interval.

7.2 Metric 30: Statistical Power Analysis

The canonical implementation delegates two-sample t-test power and sample-size calculations to `statsmodels.stats.power.tt_ind_solve_power`. The underlying planning target is Cohen's standardized effect

$$d = \frac{\mu_1 - \mu_2}{\sigma}. \tag{121}$$

The solver returns the per-group sample size n satisfying

$$\text{Power} = P\left(\text{reject } H_0 : \mu_1 = \mu_2 \mid d, \alpha, n\right) \tag{122}$$

for the requested alternative. The returned per-group sample size is $\lceil n \rceil$, while the returned total sample size follows the implementation as $\lceil 2n \rceil$. The standardized effect-size scale follows Cohen's two-sample convention [15].

95% CI (this section). an analytic design quantity; no sampling CI of its own.

7.3 Metric 31: Bias Concentration Index

For nonnegative group bias proportions $p_1, \ldots, p_K$, the implementation defines

$$BCI_{\text{bias}} = 1 - \frac{\sum_{g=1}^{K} p_g^2}{\left(\sum_{g=1}^{K} p_g\right)^2}. \tag{123}$$

If the vector is empty or sums to zero, the value is zero. For a normalized nonnegative vector, the finite-group upper bound is $1 - 1/K$, attained by an even distribution. Larger values indicate more distributed bias; smaller values indicate concentration in fewer groups.

95% CI (this section). $BCI(\mathit{concentration})$ is reported with a percentile bootstrap over the per-group bias proportions: resample with replacement $B = 1000$ times, recompute $BCI(\mathit{concentration})$, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{124}$$

7.4 Metric 32: Mutual Information Content

Let D be a demographic categorical variable and O an outcome categorical variable. In the implementation, D is expected to be encoded as nonnegative integer categories for the entropy normalization. The raw mutual information implemented via `mutual_info_score` uses Shannon mutual information [9]:

$$MIC = I(D; O) = \sum_{d,o} p(d, o) \log \frac{p(d, o)}{p(d)p(o)}. \tag{125}$$

The implementation also reports a normalized value using demographic entropy:

$$MIC_{\text{norm}} = \frac{I(D; O)}{H(D)}, \qquad H(D) = -\sum_{d} p(d) \log p(d). \tag{126}$$

If $H(D) = 0$, the normalized value is zero.

95% CI (this section). MIC is reported with a percentile bootstrap over paired (demographic, outcome) observations: resample with replacement $B = 1000$ times, recompute MIC, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{127}$$

7.5 Metric 33: Jensen-Shannon Divergence

For two nonnegative vectors normalized to probability distributions p and q, let

$$m = \frac{p+q}{2}. \tag{128}$$

The implementation returns the base-2 Jensen-Shannon divergence [11]:

$$JSD(p,q) = \frac{1}{2} D^{(2)}_{\mathrm{KL}}(p\|m) + \frac{1}{2} D^{(2)}_{\mathrm{KL}}(q\|m), \tag{129}$$

where $D^{(2)}_{\mathrm{KL}}$ uses $\log_2$. The SciPy function returns the distance $\sqrt{JSD}$, so the implementation squares that value. Both `jsd` and `jsd_distance` are reported.

95% CI (this section). JSD is reported with a percentile bootstrap over the underlying samples: resample with replacement $B = 1000$ times, recompute JSD, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{130}$$

(between aggregate distributions the interval is reported unavailable).

7.6 Metric 34: Wasserstein Distance

For one-dimensional empirical samples $P_n = \{x_1, \ldots, x_n\}$ and $Q_m = \{y_1, \ldots, y_m\}$, the implemented metric delegates to SciPy's first Wasserstein distance without explicit sample weights [16]:

$$WD(P_n, Q_m) = \inf_{\gamma \in \Gamma(P_n, Q_m)} \int_{\mathbb{R}\times\mathbb{R}} |x-y|\, d\gamma(x,y), \tag{131}$$

where P_n and Q_m are empirical distributions and $\Gamma(P_n, Q_m)$ is the set of couplings with these marginals. Equivalently, in one dimension,

$$WD(P_n, Q_m) = \int_{-\infty}^{\infty} |F_{P_n}(t) - F_{Q_m}(t)|\, dt. \tag{132}$$

95% CI (this section). WD is reported with a percentile bootstrap over each sample independently (two-sample bootstrap): resample with replacement $B = 1000$ times, recompute WD, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{133}$$

7.7 Metric 35: Network Modularity

Given an adjacency or correlation matrix A, the implementation constructs an undirected graph using $|A|$ and detects communities with greedy modularity. For total edge weight $2m$ (summing all entries A_{ij}), degree $k_i = \sum_j A_{ij}$, and community assignment c_i, Newman modularity is [17, 18]

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \mathbb{I}(c_i = c_j). \tag{134}$$

95% CI (this section). NM is reported with a percentile bootstrap over nodes (node-resampling stability bootstrap): resample with replacement $B = 1000$ times, recompute NM, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{135}$$

7.8 Metric 36: Transparency Score

For each explained decision i, the canonical appendix implementation expects three scores in $[0, 1]$: explanation quality e_i, feature importance f_i, and interpretability u_i. This score is an implementation-level aggregate for post-hoc explanation adequacy, aligned with the clinical need to expose reasons for model outputs rather than predictions alone [20]. The per-decision transparency contribution is

$$t_i = \frac{e_i + f_i + u_i}{3}. \tag{136}$$

The transparency score is the empirical mean

$$TS = \frac{1}{n}\sum_{i=1}^{n} t_i. \tag{137}$$

If no explanations are provided, the implementation returns zero.

95% CI (this section). TS is reported with a percentile bootstrap over per-decision transparency scores: resample with replacement $B = 1000$ times, recompute TS, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{138}$$

7.9 Metric 37: Robustness Certification Score

Let a_{ib} be the agreement indicator between the original prediction for case i and the prediction under perturbation batch b:

$$a_{ib} = \mathbb{I}(\hat{Y}_i = \hat{Y}^{\text{pert}}_{ib}). \tag{139}$$

For each perturbation batch,

$$r_b = \frac{1}{n}\sum_{i=1}^{n} a_{ib}. \tag{140}$$

The implemented robustness certification score is

$$RCS = \frac{1}{B}\sum_{b=1}^{B} r_b. \tag{141}$$

The implementation also reports

$$SD(RCS) = \sqrt{\frac{1}{B}\sum_{b=1}^{B}(r_b - RCS)^2}, \tag{142}$$

$$r_{\min} = \min_b r_b, \qquad r_{\max} = \max_b r_b. \tag{143}$$

The input argument ϵ is recorded in the output but is not used in the calculation itself.

8 Implemented Statistical Estimands Included to Reach 39 Derivations

95% CI (this section). RCS is reported with a percentile bootstrap over per-perturbation agreement scores: resample with replacement $B = 1000$ times, recompute RCS, and take the empirical percentiles

$$CI_{95} = \left[\hat{\theta}^*_{(0.025)},\ \hat{\theta}^*_{(0.975)}\right]. \tag{144}$$

8.1 Metric 38: Hierarchical Variance Partitioning and MAIHDA-style VPC

For a Gaussian mixed model with outcome Y_{ij} for individual i in group j, EquiMed-DSS fits a random-intercept model using the standard multilevel variance-partitioning framework [21]:

$$Y_{ij} = \beta_0 + X_{ij}^\top \beta + u_j + \varepsilon_{ij}, \qquad u_j \sim N(0, \sigma_u^2), \quad \varepsilon_{ij} \sim N(0, \sigma_e^2). \tag{145}$$

The implemented intraclass correlation from the null model is

$$ICC_{\text{HLM}} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2}. \tag{146}$$

The reported pseudo-R^2 is the proportional reduction in total variance from the null model to the full model:

$$R^2_{\text{pseudo}} = 1 - \frac{\sigma^2_{u,\text{full}} + \sigma^2_{e,\text{full}}}{\sigma^2_{u,\text{null}} + \sigma^2_{e,\text{null}}}. \tag{147}$$

For a binary outcome analysed on the logistic latent scale, the package documentation notes the MAIHDA-style variance partition coefficient

$$VPC_{\text{logit}} = \frac{\sigma_u^2}{\sigma_u^2 + \pi^2/3}. \tag{148}$$

If the mixed-model fit fails, the implementation falls back to an ANOVA-style decomposition with J groups:

$$SS_B = \sum_j n_j (\bar{Y}_j - \bar{Y})^2, \tag{149}$$

$$SS_W = \sum_i (Y_i - \bar{Y}_{g(i)})^2, \tag{150}$$

$$MS_B = \frac{SS_B}{J-1}, \qquad MS_W = \frac{SS_W}{n-J}. \tag{151}$$

With $\bar{n}$ denoting the mean group size, the fallback ICC is bounded to $[0, 1]$:

$$ICC_{\text{fallback}} = \max\left\{0, \min\left[1, \frac{MS_B - MS_W}{MS_B + (\bar{n}-1)MS_W}\right]\right\}. \tag{152}$$

8.2 Metric 39: Causal Mediation and Proportion Mediated

Let X be the treatment or exposure, M the mediator, Y the outcome, and C optional covariates. The implemented product-of-coefficients mediation uses linear regression models [22, 23]:

$$M = \alpha_0 + \alpha_1 X + C^\top \alpha_C + \varepsilon_M, \tag{153}$$

$$Y = \beta_0 + \beta_1 X + \beta_2 M + C^\top \beta_C + \varepsilon_Y. \tag{154}$$

A separate total-effect model is

$$Y = \tau_0 + \tau_1 X + C^\top \tau_C + \varepsilon_T. \tag{155}$$

The indirect, direct, and total effects are

$$IE = \alpha_1 \beta_2, \tag{156}$$
$$DE = \beta_1, \tag{157}$$
$$TE = \tau_1. \tag{158}$$

The implemented proportion mediated is

$$PM = \frac{IE}{TE}, \tag{159}$$

with $PM = 0$ if $|TE| \leq 10^{-10}$. The indirect-effect confidence interval is a percentile bootstrap over the product $\alpha_1^* \beta_2^*$. The Sobel test implemented in the same class is

$$SE_{\text{Sobel}} = \sqrt{\alpha_1^2 SE(\beta_2)^2 + \beta_2^2 SE(\alpha_1)^2}, \tag{160}$$
$$z_{\text{Sobel}} = \frac{\alpha_1 \beta_2}{SE_{\text{Sobel}}}, \tag{161}$$
$$p = 2\left[1 - \Phi(|z_{\text{Sobel}}|)\right]. \tag{162}$$

References

[1] Shrout PE, Fleiss JL. Intraclass correlations: uses in assessing rater reliability. *Psychological Bulletin*. 1979;86(2):420-428. doi:10.1037/0033-2909.86.2.420.

[2] Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet*. 1986;327(8476):307-310. doi:10.1016/S0140-6736(86)90837-8.

[3] Wilson EB. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*. 1927;22(158):209-212. doi:10.1080/01621459.1927.10502953.

[4] Equal Employment Opportunity Commission, Civil Service Commission, Department of Labor, Department of Justice. Uniform Guidelines on Employee Selection Procedures. *Federal Register*. 1978;43(166):38290-38315.

[5] Gini C. *Variabilita e Mutabilita*. Bologna: Tipografia di Paolo Cuppini; 1912.

[6] Shewhart WA. *Economic Control of Quality of Manufactured Product*. New York: D. Van Nostrand Company; 1931.

[7] Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. In: *Proceedings of the 34th International Conference on Machine Learning*. PMLR. 2017;70:1321-1330.

[8] Kusner MJ, Loftus J, Russell C, Silva R. Counterfactual fairness. In: *Advances in Neural Information Processing Systems*. 2017;30.

[9] Shannon CE. A mathematical theory of communication. *The Bell System Technical Journal*. 1948;27(3):379-423, 27(4):623-656.

[10] Kullback S, Leibler RA. On information and sufficiency. *Annals of Mathematical Statistics.* 1951;22(1):79-86. doi:10.1214/aoms/1177729694.

[11] Lin J. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory.* 1991;37(1):145-151. doi:10.1109/18.61115.

[12] Gibbs AL, Su FE. On choosing and bounding probability metrics. *International Statistical Review.* 2002;70(3):419-435. doi:10.1111/j.1751-5823.2002.tb00178.x.

[13] Efron B. Bootstrap methods: another look at the jackknife. *Annals of Statistics.* 1979;7(1):1-26. doi:10.1214/aos/1176344552.

[14] Efron B, Tibshirani RJ. *An Introduction to the Bootstrap.* New York: Chapman and Hall/CRC; 1993.

[15] Cohen J. *Statistical Power Analysis for the Behavioral Sciences.* 2nd ed. Hillsdale: Lawrence Erlbaum Associates; 1988.

[16] Villani C. *Optimal Transport: Old and New.* Berlin: Springer; 2009.

[17] Newman MEJ. Modularity and community structure in networks. *Proceedings of the National Academy of Sciences of the United States of America.* 2006;103(23):8577-8582. doi:10.1073/pnas.0601602103.

[18] Clauset A, Newman MEJ, Moore C. Finding community structure in very large networks. *Physical Review E.* 2004;70:066111. doi:10.1103/PhysRevE.70.066111.

[19] Shapley LS. A value for n-person games. In: Kuhn HW, Tucker AW, eds. *Contributions to the Theory of Games II.* Princeton: Princeton University Press; 1953:307-317.

[20] Ribeiro MT, Singh S, Guestrin C. "Why should I trust you?": explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.* 2016:1135-1144. doi:10.1145/2939672.2939778.

[21] Raudenbush SW, Bryk AS. *Hierarchical Linear Models: Applications and Data Analysis Methods.* 2nd ed. Thousand Oaks: Sage Publications; 2002.

[22] Sobel ME. Asymptotic confidence intervals for indirect effects in structural equation models. *Sociological Methodology.* 1982;13:290-312. doi:10.2307/270723.

[23] Imai K, Keele L, Tingley D. A general approach to causal mediation analysis. *Psychological Methods.* 2010;15(4):309-334. doi:10.1037/a0020761.

Implementation Notes

- BEMI is total variation distance, not one minus a correlation.
- GRBI is directed KL divergence in nats, not a symmetric distance.
- JSD is reported as divergence in base 2. The corresponding distance is also exposed as the square root.
- ECS is implemented as cosine distance, so lower values mean greater consistency.

- HAFG in `domain2` is a two-group count-based metric normalized by the larger harm. wHAFG in `domain5` is a per-sample, severity-weighted multi-group generalization.
- Bootstrap confidence intervals in the canonical appendix implementation use percentile intervals only.
- The prior PDF derivations for SPG, CHR, IVI, GRI, ICE, wHAFG, LDDI, and related metrics are broadly consistent with the implementation, but this file uses the exact implemented details where the older derivations were more general.

9 Uncertainty Quantification for Every Metric (v1.9.5)

From v1.9.5 a metric is never reported as a bare point estimate. Each estimate $\hat{\theta}$ carries a 95% confidence interval and, where a pre-specified null or acceptability threshold exists, a p-value. Three master estimators cover every metric; the table tags each metric with the one it uses, and the proportion metrics (CHR, IVI, DFR) carry their CI and p-value directly in the result dict. Let $z = z_{1-\alpha/2} = \Phi^{-1}(1-\alpha/2)$ (so $z = 1.96$ at 95%), Φ the standard-normal CDF, and B, P the bootstrap and permutation resample counts.

9.1 A. Binomial-proportion metrics: Wilson interval and score test

For a proportion $\hat{p} = k/n$ (CHR, IVI, DFR, Safety Violation Rate, Audit Traceability, Counterfactual Unfairness, Transparency, Robustness Certification), the library returns the Wilson score interval and a one-sided score test against an acceptability threshold p_0 (default 0.05):

$$\mathrm{CI}_{1-\alpha} = \frac{1}{1+z^2/n}\left[\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}\right], \tag{163}$$

$$Z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}, \qquad p = 1-\Phi(Z) \ \ (\text{alt. } \hat{p} > p_0). \tag{164}$$

9.2 B. Sample statistics: bootstrap (with cluster option)

For $\hat{\theta} = T(x_1, \ldots, x_n)$ a mean, distance, entropy, Gini, total-variation distance, or divergence (ECS, ICC, SPG, ICE, CPS, SRPI, LDDI, REG, CIDR, UQG, GRBI, BEMI, GCC, ISFV, MIC, JSD, Wasserstein), the percentile bootstrap uses B resamples $x^{*(b)}$ drawn with replacement:

$$\hat{\theta}^{*(b)} = T(x^{*(b)}), \qquad \mathrm{CI}_{1-\alpha} = \left[\hat{\theta}^*_{(\alpha/2)},\ \hat{\theta}^*_{(1-\alpha/2)}\right], \tag{165}$$

the empirical quantiles of the bootstrap distribution, with $\widehat{\mathrm{se}} = \mathrm{sd}(\hat{\theta}^{*(1)}, \ldots, \hat{\theta}^{*(B)})$. For clustered observations (repeated evaluations per patient/visit), the *cluster bootstrap* resamples whole clusters $g \in \{1, \ldots, G\}$ rather than rows, $\hat{\theta}^{*(b)} = T(\bigcup_j x_{g_j^*})$ with $\{g_j^*\} \sim \mathrm{Unif}\{1, \ldots, G\}$, widening the interval to reflect within-cluster correlation (the interval reported for the per-patient proportions). Any metric can be wrapped with `inference.bootstrap_metric`.

9.3 C. Between-group gaps: permutation test

For a fairness gap $\Delta = \hat{\theta}_A - \hat{\theta}_B$ (HER, HAFG/wHAFG, IBS, HSSF, between-group Theil, Temporal Fairness Drift), the permutation test shuffles group labels P times and reports the add-one p-value

$$p = \frac{1 + \#\{b:\ |\Delta^{\pi_b}| \geq |\Delta_{\mathrm{obs}}|\}}{1+P}, \tag{166}$$

which is never exactly zero.

9.4 Worked instances (manuscript metrics)

$\widehat{\mathrm{CHR}} = 274/285 = 0.961$, Wilson 95% CI $[0.932, 0.978]$, $p < 0.001$ vs $p_0 = 0.05$; $\widehat{\mathrm{IVI}} = 36/132 = 0.273$, 95% CI $[0.204, 0.354]$, $p < 0.001$; per-patient accuracy 0.263, cluster (by-visit) bootstrap 95% CI $[0.235, 0.292]$, entirely below the 0.443 baseline; $\mathrm{BEMI} = \frac{1}{2}\sum_r |e_r - b_r| = 0.67$ with a record-level bootstrap CI (estimator B).